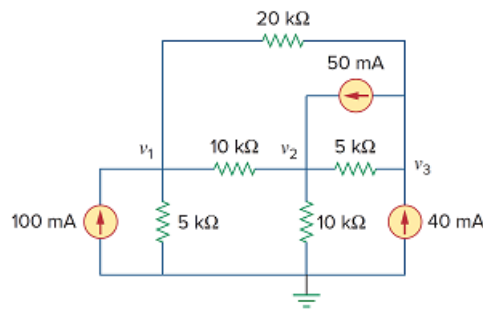


Problem

For the circuit shown in Fig. 3.113, write the node-voltage equations by inspection.

Figure 3.113:



Step-by-step solution

Step 1 of 8

Refer to Figure 3.113 in the text book.

If a circuit has N non-reference nodes, the node-voltage equations can be expressed in terms of conductance:

$$\mathbf{G}\mathbf{v} = \mathbf{i}$$

$$\begin{pmatrix} G_{11} & \cdots & G_{1N} \\ \vdots & \ddots & \vdots \\ G_{N1} & \cdots & G_{NN} \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_N \end{pmatrix} = \begin{pmatrix} i_1 \\ \vdots \\ i_N \end{pmatrix}$$

Where,

Sum of the conductance connected to node k is G_{kk} .

Negative of the sum of the conductance directly connecting nodes k and j , $k \neq j$ is, $G_{kj} = G_{jk}$

Unknown voltage at node k is i_k .

Sum of all independent current sources directly connected to node k , with currents entering the node treated as positive is v_k .

Step 2 of 8

Each of the diagonal terms is the sum of the conductance in the related to node, while each of the off-diagonal terms is the negative of the conductance common to nodes.

The circuit has three non-reference nodes, so there are three node equations. This implies that the size of the conductance matrix **G**, is 3 by 3.

Calculate the diagonal terms of **G** in Siemens.

Calculate the value of G_{11} .

$$\begin{aligned} G_{11} &= \frac{1}{5 \text{ k}\Omega} + \frac{1}{10 \text{ k}\Omega} + \frac{1}{20 \text{ k}\Omega} \\ &= \frac{4+2+1}{20 \text{ k}\Omega} \\ &= \frac{7}{20 \text{ k}\Omega} \\ &= 0.35 \text{ mS} \end{aligned}$$

Therefore, the value of the G_{11} is 0.35 mS.

Step 3 of 8

Calculate the value of G_{22} .

$$\begin{aligned} G_{22} &= \frac{1}{5 \text{ k}\Omega} + \frac{1}{10 \text{ k}\Omega} + \frac{1}{10 \text{ k}\Omega} \\ &= \frac{2+1+1}{10 \text{ k}\Omega} \\ &= \frac{4}{10 \text{ k}\Omega} \\ &= 0.4 \text{ mS} \end{aligned}$$

Therefore, the value of the G_{22} is 0.4 mS.

Step 4 of 8

Calculate the value of G_{33} .

$$\begin{aligned} G_{33} &= \frac{1}{5 \text{ k}\Omega} + \frac{1}{20 \text{ k}\Omega} \\ &= \frac{4+1}{20 \text{ k}\Omega} \\ &= \frac{5}{20 \text{ k}\Omega} \\ &= 0.25 \text{ mS} \end{aligned}$$

Therefore, the value of the G_{33} is 0.25 mS.

Step 5 of 8

Calculation of non-diagonal terms:

Calculate the value of G_{12} .

$$\begin{aligned} G_{12} &= -\frac{1}{10 \text{ k}\Omega} \\ &= -\frac{1}{10 \text{ k}\Omega} \\ &= -0.1 \text{ mS} \end{aligned}$$

Therefore, the value of the G_{12} is -0.1 mS .

Since G_{21} is equal to G_{12} ,

The value of the G_{21} is equal to -0.1 mS .

Step 6 of 8

Calculate the value of G_{13} .

$$\begin{aligned} G_{13} &= -\frac{1}{20 \text{ k}\Omega} \\ &= -0.05 \text{ mS} \end{aligned}$$

Therefore, the value of the G_{13} is -0.05 mS .

Since G_{31} is equal to G_{13} ,

The value of the G_{31} is equal to -0.05 mS .

Step 7 of 8

Calculate the value of G_{23} .

$$\begin{aligned} G_{23} &= -\frac{1}{5 \text{ k}\Omega} \\ &= -0.2 \text{ mS} \end{aligned}$$

Therefore, the value of the G_{23} is -0.2 mS .

Since G_{32} is equal to G_{23} ,

The value of the G_{23} is equal to -0.2 mS .

Step 8 of 8

Write node voltage equations in matrix representation.

$$\begin{bmatrix} G_{11} & G_{12} & G_{13} \\ G_{21} & G_{22} & G_{23} \\ G_{31} & G_{32} & G_{33} \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix}$$

Substitute -0.05 mS for G_{13} and G_{31} , -0.2 mS for G_{23} and G_{32} , -0.1 mS for G_{21} and G_{12} , 0.35 mS for G_{11} , 0.4 mS for G_{22} , 0.25 mS for G_{33} , 100 mA for i_1 , 50 mA for i_2 and $(40 \text{ mA} - 50 \text{ mA})$ for i_3 .

$$\begin{bmatrix} 0.35 \text{ mS} & -0.1 \text{ mS} & -0.05 \text{ mS} \\ -0.1 \text{ mS} & 0.4 \text{ mS} & -0.2 \text{ mS} \\ -0.05 \text{ mS} & -0.2 \text{ mS} & 0.25 \text{ mS} \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 100 \text{ mA} \\ 50 \text{ mA} \\ 40 \text{ mA} - 50 \text{ mA} \end{bmatrix}$$

$$\begin{bmatrix} 0.35 \text{ mS} & -0.1 \text{ mS} & -0.05 \text{ mS} \\ -0.1 \text{ mS} & 0.4 \text{ mS} & -0.2 \text{ mS} \\ -0.05 \text{ mS} & -0.2 \text{ mS} & 0.25 \text{ mS} \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 100 \text{ mA} \\ 50 \text{ mA} \\ -10 \text{ mA} \end{bmatrix}$$

$$\begin{bmatrix} 0.35 & -0.1 & -0.05 \\ -0.1 & 0.4 & -0.2 \\ -0.05 & -0.2 & 0.25 \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 100 \\ 50 \\ -10 \end{bmatrix}$$

Therefore, the set of node voltage equations for the given circuit is,

$$\boxed{\begin{bmatrix} 0.35 & -0.1 & -0.05 \\ -0.1 & 0.4 & -0.2 \\ -0.05 & -0.2 & 0.25 \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 100 \\ 50 \\ -10 \end{bmatrix}}$$